



Export-Oriented Agricultural Technology, Employment, and Economic Growth Among Smallholder Farmers in Bangladesh

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Abstract: The poverty-reducing potential of agricultural employment, in conjunction with export-oriented technological innovation, was critically examined within the context of Bangladesh's agrarian economy. Particular emphasis was placed on the extent to which agricultural research and development outputs—including technology transfer, patent generation, and research dissemination—contribute to gross domestic product per capita growth, a key proxy for economic development. It was demonstrated that export-oriented agricultural technologies were significantly associated with economic growth. In contrast, general government expenditure on agricultural research and development, when implemented without clear innovation-oriented objectives, was shown to exert a limited effect on economic growth. The findings suggest that the effectiveness of research and development investment is contingent not on scale alone, but on the capacity to generate scalable, market-oriented technological outputs. Moreover, structural dimensions of poverty were addressed by illustrating how a technology-driven and employment-intensive agricultural system functioned as a critical mechanism for inclusive development. It was further observed that poverty reduction was deeply embedded within broader institutional, economic, and policy frameworks, necessitating coordinated interventions that integrate technological advancement with equitable resource distribution. The analysis underscores the importance of targeted policy design aimed at fostering innovation ecosystems that prioritize export competitiveness, rural employment generation, and sustainable income growth. Such an approach is argued to facilitate a transition toward a more resilient, inclusive, and self-sustaining development trajectory in Bangladesh.

Keywords: Agricultural employment; Export-oriented innovation; Poverty reduction; Agricultural research and development; Economic growth; Structural poverty

1. Introduction

Progress in economic growth and sustainable development in Bangladesh increasingly hinges on modernization within agricultural employment and the export of agricultural technologies (Siddique, 2025). As an overwhelmingly agrarian economy, Bangladesh derives livelihoods and income for over two-thirds of its population from agriculture. However, persistent reliance on outdated farming practices, limited access to modern technology, and underdeveloped agricultural value chains continue to inhibit the sector's full potential. Achieving long-term, inclusive economic growth thus requires leveraging agricultural employment and advancing technology-driven exports to reduce poverty and foster development (Pinto et al., 2025).

Agricultural research and development, particularly when innovations are effectively commercialized, can transform the rural economy by increasing productivity, improving food security, and raising incomes (Fei et al., 2025). Innovations such as advanced farm machinery, improved irrigation systems, high-yield crop varieties, and export-oriented post-harvest technologies exemplify the kinds of breakthroughs necessary for meaningful impact. Within the framework of the Solow growth model, technological progress is the primary driver of long-run economic growth. Applied to agriculture, this implies that exports of technology-based products can increase total factor productivity and enhance financial performance. Findings suggest that a 1% increase in research and

development expenditure can raise economic growth by 0.13%, emphasizing that strategic investment in innovation yields measurable macroeconomic gains.

Innovation also plays a vital role in shaping market dynamics and creating new economic opportunities (Corvello, 2025). It enables the development of novel products and practices that can open access to international markets, diversify export portfolios, and generate foreign exchange earnings. Research by the National Bureau of Economic Research underscores that over 50% of economic growth in developed nations, such as the United States, is attributable to innovation. Extrapolating from this evidence, agricultural innovation geared toward exports in Bangladesh holds the potential to generate rural employment, stimulate community-level growth, and reduce poverty.

Nevertheless, several challenges persist. Bangladesh faces infrastructural and institutional constraints in its research and development sector, compounded by limited financial resources and a shortage of skilled human capital, especially in rural areas (Fei et al., 2025). Private sector engagement remains low due to high perceived risks and long payback periods, while budgetary limitations constrain public investment. Furthermore, existing literature shows mixed results regarding the link between research and development spending and economic growth. Policy discourse in Bangladesh has increasingly emphasized the role of agricultural technology exports as a strategic route to sustainable economic development (Datta et al., 2024). However, empirical studies investigating the causal connections among agricultural employment, technology export, and growth remain scarce. Addressing this research gap, the present study aims to explore the relationship between farm employment and technology generation and their combined impact on poverty alleviation and sustainable growth in Bangladesh.

To achieve this objective, the study examines key variables, including agricultural labor, agricultural technology exports, research and development investment, and innovation outputs (patents and commercial technologies). The dependent variable, economic impact, is measured by gross domestic product per capita growth. The study proposes several hypotheses for empirical testing intended to inform policy and planning decisions (Li & Zhou, 2024). In doing so, it adopts a structural lens on poverty, emphasizing how entrenched dominant economic, political, and institutional arrangements maintain inequalities. By focusing on innovation-led, employment-intensive agricultural development, the research seeks to identify pathways that can dismantle systemic poverty and promote inclusive growth.

This study formulates the following research hypotheses:

H1: Government expenditure on agricultural research and development has a significant effect on sustainable economic growth in Bangladesh.

H2: The development and dissemination of agricultural research outputs (such as peer-reviewed publications and knowledge sharing) positively impact gross domestic product per capita growth by enhancing innovation and productivity in the agricultural sector.

H3: The application and registration of agricultural technology patents have a positive and statistically significant effect on economic growth in Bangladesh.

H4: The export of value-added agricultural technologies has a statistically significant and positive impact on poverty reduction and sustainable economic growth in Bangladesh.

2. Literature Review

2.1 Theoretical Framework

This study employs several theoretical models to examine the linkage between agricultural employment, export technology, and sustainable economic growth in Bangladesh. A particularly pertinent model in this regard is the endogenous growth theory, which posits that endogenous factors such as innovation, human capital, and sector-specific progress, especially in agriculture, can lead to long-run economic growth (Abdulle et al., 2025). From this perspective, investments in agricultural research and technological development can result in new techniques, practices, and tools that increase productivity, improve food security, and stimulate employment in rural areas.

Within this context, agriculture serves a dual purpose: it provides subsistence for a significant portion of the population, particularly in rural areas (Lowder et al., 2025). It acts as a driver of inclusive economic growth. The modernization and dissemination of traditional agro-practices through agricultural research publications and demonstrations facilitate knowledge transfer. Advances of usable farm technologies such as improved seed varieties, sustainable irrigation methods, and enhanced mechanization can elevate both the quantity and quality of output across the sector.

Agricultural patents and innovations reflect a region's technological capabilities and skill levels. The export of agricultural technology products, including processing equipment, bio-fertilizers, and post-harvest tools, demonstrates a nation's competence in competing in global markets (El Malahi et al., 2025). These technologies not only support domestic agriculture but also contribute to foreign exchange earnings and strengthen international relationships. Furthermore, the integration of artificial intelligence in agriculture has shown promising results. For instance, artificial intelligence-enabled agricultural information systems in Bangladesh have empowered farmers

by providing soil testing, tailored advice, and market connectivity, leading to increased yields and reduced input costs. In aggregate, agricultural labor, innovation, and export-oriented technology are pivotal elements conducive to productivity growth, income generation, and rural transformation. This interaction underpins the rationale for promoting agriculture as an engine of poverty reduction and sustainable economic development in Bangladesh (Datta et al., 2024). Addressing structural causes of poverty and inequality requires a focus on empowering marginalized communities through access to technology, education, and market opportunities, thereby challenging dominant economic, political, cultural, and social institutions.

A substantial body of literature highlights the critical role of agricultural innovation in driving economic growth, particularly in developing economies. According to endogenous growth theory, investments in research and development, human capital, and technological advancement contribute to long-term productivity gains. Empirical studies show that agricultural innovation such as improved seed varieties, mechanization, and digital farming technologies enhances efficiency, increases output, and supports income growth. Research indicates that knowledge creation, often measured through scientific publications and research outputs, plays a vital role in facilitating innovation and technology diffusion. Similarly, patent activity reflects the application of research into commercially viable technologies, which further strengthens economic performance. The export of technology-intensive agricultural products has emerged as a key driver of economic development. Studies suggest that countries integrating innovation into export markets benefit from increased foreign exchange earnings, improved competitiveness, and structural transformation. High-value agricultural exports, including processed goods and technology-based inputs, contribute not only to gross domestic product growth but also to the diversification of the economy. However, the effectiveness of export-led growth depends on institutional quality, infrastructure, and the ability to commercialize innovation. Without these supporting factors, the impact of technology exports on economic growth may remain limited or inconsistent.

2.2 Empirical Literature

An expanding body of evidence indicates that advancements in agriculture and technology exports can substantially impact a nation's economic growth and development (Sharma et al., 2025). Investments in sector-specific innovations have demonstrated high returns, particularly in terms of per capita income growth. In Bangladesh, targeted investments in agricultural research and rural labor force training have led to increased productivity, enhanced food security, and job creation. In economies where agriculture constitutes a significant component, even modest increases in spending on agricultural technologies have yielded noticeable improvements in gross domestic product per capita (Suri & Udry, 2022). Technological innovations in agrarian production, such as precision agriculture, organic farming, and cold storage solutions, have been instrumental in promoting productivity and reducing post-harvest losses, thereby increasing marketable surplus and farmer returns.

The development of agricultural employment, when coupled with technological and institutional support, has proven effective in poverty alleviation, especially in developing nations. The introduction of mechanized farming, mobile-based farming advisories, and micro-irrigation solutions has enhanced labor productivity and alleviated the drudgery associated with traditional agriculture (Gautam et al., 2023). These advancements empower rural communities to participate more effectively in economic activities, thereby improving living standards and consumption levels. Countries that have successfully developed and commercialized agricultural technologies have experienced significant effects on overall economic growth (Pillai, 2025). The development and export of value-added farm products and farming technologies not only diversify the economic structure but also enhance innovation capacity and global competitiveness. However, the positive effects of these exports are more pronounced in countries with well-developed infrastructure, trained human capital, and efficient market access.

Nonetheless, research indicates that the relationship between agricultural innovation expenditure and economic growth is not uniformly positive or linear (Waheed et al., 2025). In contexts characterized by weak institutions, poor coordination, and inadequate market linkages, high investments may yield limited impact. Without an enabling environment, adequate training, or infrastructure, the potential benefits of technological and agricultural development can be significantly diminished (Kabato et al., 2025). Therefore, countries like Bangladesh must tailor their agricultural policies to local needs, capabilities, and opportunities. Establishing appropriate institutional frameworks, providing farmers with access to advanced tools and knowledge, and promoting export-oriented agricultural innovations are crucial steps toward poverty alleviation and sustainable economic development. Agricultural development plays a central role in poverty alleviation, particularly in rural economies like Bangladesh. Increased productivity through technological adoption can raise farm incomes, create employment opportunities, and improve living standards. Studies show that when innovation is combined with access to markets, extension services, and institutional support, it significantly contributes to reducing poverty and inequality. However, structural barriers such as limited access to finance, weak institutions, and low human capital can constrain these benefits.

3. Methodology

This research employs a quantitative approach to analyze the effects of agricultural employment and export technology on poverty alleviation and sustainable economic development in Bangladesh. Quantitative methods enable researchers to systematically collect and analyze numerical data to identify patterns, associations, and significance between factors related to agriculture and economic issues. The study is based on secondary data sourced from reputable international and national institutions that provide macroeconomic, employment, and agricultural technology export statistics. To examine the connection between farm employment and export technology to gross domestic product, the research utilizes multiple regression analysis. Specifically, it assesses the impact of independent variables such as the share of farm employment and the level of technology transferred through agricultural product exports on dependent variables like gross domestic product per capita and poverty reduction levels. The elasticity with which real gross domestic product responds to employment and technological exports is calculated to determine the statistical significance and strength of these predictive variables. This data-driven approach contributes to an evidence-based understanding of how investments in agriculture and technology-led solutions can serve as strategic instruments for facilitating sustainable development and poverty reduction in Bangladesh. Recent studies support this methodology. For instance, the autoregressive distributed lag method has been employed to assess the contribution of agricultural subsectors to Bangladesh's economic growth, finding a strong long-term association between these subsectors and economic development. Similarly, time series analysis has been used to reveal a positive impact of agricultural credit on gross domestic product growth in Bangladesh. These studies underscore the importance of sector-specific investments and the use of robust quantitative methods to inform policy decisions aimed at poverty alleviation and sustainable economic growth.

3.1 Model Specification

Theoretical perspectives and empirical evidence increasingly indicate that greater investments in agriculture and export-focused technology can significantly contribute to economic growth and poverty reduction, especially in developing economies like Bangladesh. The Solow growth model and endogenous growth theory provide the foundation for understanding how productivity-enhancing investments and technological advancement can lead to long-term improvements in living standards. These theories suggest that knowledge accumulation and innovation in the agricultural sector can serve as engines of sustained development. Building on this premise, this study develops a conceptual model that links farm employment and the export of technology-intensive agricultural products to economic performance and poverty alleviation.

The model considers four independent variables: the agricultural employment rate, technology-based agricultural exports, rural household net income, and agricultural machinery availability. The dependent variable growth of gross domestic product per capita is employed as a proxy for sustainable economic development. This framework allows for the investigation of how structural changes in rural labor allocation, innovation in agri-export systems, and mechanization affect poverty and inequality in rural Bangladesh. Empirical evidence from recent studies shows that agriculture-led innovation, when integrated with improved export systems and supported by rural income-enhancing policies, leads to measurable improvements in household welfare and national economic indicators. By drawing on these relationships, the model provides critical insight into how dominant economic, institutional, and technological structures shape poverty and inequality.

The model is specified as follows:

$$Y = \beta_0 + \beta_1(X_1) + \beta_2(X_2) + \beta_3(X_3) + \beta_4(X_4) + \varepsilon \quad (1)$$

where, Y represents the gross domestic product per capita growth, reflecting the percentage change in national income per individual over time; β_0 is the intercept term, indicating the baseline level of gross domestic product per capita growth when all explanatory variables are zero; and X_1 denotes the agricultural employment rate, which measures the proportion of the labor force engaged in agriculture. A higher employment rate in agriculture, especially when linked to productivity gains, is expected to influence economic growth and reduce rural poverty positively. X_2 refers to the value of agricultural technology exports, particularly processed or mechanized products, as a percentage of total exports. This variable indicates the country's capacity to commercialize agriculture through innovation and compete in global markets. X_3 represents the rural household income growth, which reflects the economic well-being of the agricultural labor force. An increase in rural income is likely to lead to greater consumption, investment in local economies, and overall poverty reduction. X_4 represents the level of agricultural mechanization, measured by the rate of adoption of tools, machinery, or digital farming technologies in rural areas. Mechanization enhances productivity, reduces labor intensity, and increases efficiency in the sector. ε is the error term accounting for the effects of unobserved variables not captured in the model. Table 1 shows the variables and their definitions and data sources.

Table 1 interprets the calculated coefficients of the independent variables. All the coefficients (β_1 , β_2 , β_3 , and β_4) are marginal effects of each of the independent variables on gross domestic product per capita growth. A priori, β_1 is expected to be positive, indicating that increased agricultural employment contributes positively to gross domestic product per capita growth when supported by appropriate policy interventions and workforce training. In addition, β_2 is expected to be positive, as increasing exports of value-added agriculture are indicative of technological progress and international competitiveness. The parameter β_3 is expected to be strongly positive in gross domestic product per capita growth since higher household income generally means improved agricultural output and reduced poverty. Finally, β_4 should be positive as well, underlining the role of the adoption of technology and innovation in increasing the productivity and sustainability of agriculture.

Table 1. Variables, definitions, and data sources

Variable	Symbol	Measurement	Proxy Interpretation
Gross domestic product (GDP) per capita growth	γ	Annual percentage growth rate of real GDP per capita	Economic development indicator
Agricultural employment rate	X_1	Percentage of total labor force engaged in the agricultural sector	Labor allocation in agriculture, rural labor structure
Agricultural technology exports	X_2	Value of agricultural technology and processed agricultural product exports, as a percentage of total merchandise exports	Commercialization capacity of agricultural innovation, global market competitiveness
Rural household income growth	X_3	Annual percentage growth rate of rural household disposable income per capita	Economic well-being of agricultural labor force, the effect of rural poverty reduction
Agricultural mechanization level	X_4	Adoption rate of agricultural machinery, equipment and digital farming technologies in rural areas	Agricultural productivity enhancement, production efficiency improvement

The ordinary least squares method is employed to estimate the regression model. In this way, the sum of squares of the differences (residuals) between observed and predicted values is minimized to find the best fit line in the data. To assess the relationship between agricultural employment and export technology and economic growth, the expected signs and statistical significance of the coefficients of agricultural employment and export technology are examined in the model. Specifically, the study tests the hypothesis that agricultural employment and export technology significantly affect economic growth in Bangladesh. Before initiating the regression analysis, standard diagnostic tests, including normality, multicollinearity, and linearity checks, are conducted to validate the model. These procedures ensure that the fundamental assumptions of linear regression are satisfied and that the variables exhibit appropriate relationships for reliable estimation. Verifying these assumptions is essential in avoiding biased coefficients or spurious relationships, particularly when analyzing macroeconomic and sector-specific data in a developing country context.

This study adopts a quantitative research design to examine the impact of agricultural innovation and export-oriented technology on economic growth in Bangladesh. The analysis is based on secondary time-series data collected from internationally recognized sources, including the World Bank and the World Intellectual Property Organization. To ensure the validity and reliability of the regression results, several standard econometric tests are performed. Stationarity is examined using unit root tests such as the augmented Dickey–Fuller test to avoid spurious relationships. Multicollinearity among independent variables is assessed using the variance inflation factor, while heteroskedasticity is tested using the Breusch–Pagan test. A Wald test is applied to verify the linearity of the model, and residual normality is also checked. The results of these diagnostic tests indicate that the assumptions of the ordinary least squares model are reasonably satisfied, suggesting that the estimated coefficients are robust and the model provides a reliable basis for analyzing the relationship between innovation and economic growth in Bangladesh.

This analytical model offers a structured approach to understanding how exports of agriculture and technology-based industries can serve not only as engines of economic growth but also as vehicles for long-term poverty reduction and sustainable income generation in Bangladesh. Recent research emphasizes the role of export diversification, institutional quality, and technology adoption in reducing inequality and achieving inclusive development. Moreover, empirical evidence suggests that export-led agricultural growth, especially when complemented by rural investment, extension services, and digital innovation, can have pronounced effects on poverty alleviation, particularly in underdeveloped regions. This aligns with the structuralist view that poverty and inequality are sustained by systemic economic and institutional exclusion and that inclusive trade and innovation policies are needed to break these cycles.

Table 2 shows the association between different industries, agricultural employment, and export technology for poverty alleviation and sustainable economic growth in Bangladesh. These variables are also important for

assessing the effect of these factors on economic development concerning agricultural employment and technology exports.

Table 2. Collinearity test (coefficient diagnostic test)

Variable	Coefficient	Uncentered	Centered
C	1,608.53	18.62	NA
Agricultural employment	384.05	3.97	2.18
Agricultural export technology	15,801.25	21.88	6.22
Technological innovation export	5.07	3.80	1.45
High-tech exports	36.19	65.25	6.49

Note: NA = Not applicable.

Table 3 delineates the results of the linearity test to examine the association between agricultural employment and export technology in social and economic aspects of development that influence sustainable economic growth and poverty alleviation in Bangladesh. The significance of the variables (confirmed by the Wald test) gives an overall picture of the relationship through which export technology and agricultural employment are linked to achieving long-run economic development in the country.

Table 3. Linearity test

Wald Test			
Test Statistic	Value	Degrees of Freedom	Probability
F-statistic	103.19	(4, 16)	<0.01
Chi-square	412.75	4	<0.01
Null hypothesis: $C(2) = C(3) = C(4) = C(5) = 0$			
Null Hypothesis Summary			
Variable	Value	Standard Error	
C (2)	0.04	0.01	
C (3)	20.89	6.02	
C (4)	5.74	2.25	
C (5)	133.94	125.70	

4. Results and Discussion

Descriptive statistics of predictor and response variables employed in the study are presented in the table below. The maximal values show that the dependent variables have changed over time. In any case, all the variables are improved in some degree (with differences in the degree and the scale of the improvement). While there is some variation for a few of these factors, the pattern of results indicates an overall trend of positive change over time. More specifically, as shown on the left side, employment in agriculture and exports of agricultural technology have been on the rise since the base year of the research. Similarly, the patterns of farm output and export growth also exhibit a positive ascending trend, albeit not completely smooth.

Table 4 presents descriptive statistics of select variables of interest, namely those associated with agricultural employment and export technology and that play a key part in shaping poverty reduction and sustainable economic growth in Bangladesh. The data set consisting of agricultural patents, spending on research, the growth rate of gross domestic product per capita, citable journal articles, and technology exports provides information on the current state of these correlates.

Table 4. Descriptive analysis of variables in the study

Variable	Number of Observations	Minimum	Maximum	Mean	Standard Deviation
Number of patent applications	21	0.00	18.00	6.33	5.09
Research and development expenditure	21	0.01	0.60	0.29	0.19
Gross domestic product per capita growth	21	255.10	834.96	505.83	197.21
Number of citable journal articles	21	261.00	8,876.00	2,005.38	2,270.06
Medium and high-tech export (value added)	21	6.26	16.70	11.84	4.03

This regression model (Table 5) explores how factors such as agricultural innovation, export technology and research and development expenditures affect the increase in the gross domestic product per capita. The findings reveal important associations of the number of citable journal articles, patent applications, and medium/high-tech exports with economic growth, which is essential to sustain economic growth and reduce poverty in Bangladesh.

Table 5. Predictors' coefficients of the regression model

Variable	β	Standard Error	t-Statistic	p-Value
C	109.72	40.11	2.74	0.01
Number of citable journal articles	0.04	0.01	5.89	<0.01*
Research and development expenditure (as a percentage of gross domestic product)	133.94	125.70	1.07	0.30
Number of patent applications	5.74	2.26	2.55	0.02
Medium and high-tech export (value added)	20.89	6.02	3.47	0.00*

Note: * indicates that p is significant ($p < 0.05$).

4.1 Linear Estimations (Multiple Linear Regression Analysis)

The associations between the coefficients and predictor variables with economic growth are presented in Table 5. Agricultural employment is positively related to economic growth and statistically significant at 1% ($\beta = 0.036486$, $t = 5.888918$, $p < 0.01$). This suggests that an increase in agricultural employment by one unit leads to a 0.036486% growth in gross domestic product on average. In contrast, the transfer of agricultural technology is not a significant contributor to economic growth ($\beta = 133.9384$, $t = 1.065514$, $p = 0.3025$). This result implies that there is insufficient evidence to suggest that, *ceteris paribus*, an increase in agricultural technology exports directly leads to significant economic growth. Although technology-oriented exports do show both positive and negligible relationships with economic growth, the relationship is complex. It depends on the type of technologies exported, market demand, and economic conditions.

Agricultural research and development expenditure is another key variable examined in this study. As presented in Table 5, the coefficients of farming patents ($\beta = 5.741583$, $t = 2.549996$, $p = 0.0214$) and export of value-added agricultural goods ($\beta = 20.88898$, $t = 3.472164$, $p = 0.0031$) suggest that each unit increase in patents and the export of value-added agricultural goods leads to 5.74% and 20.89% growth in per capita gross domestic product, respectively. The small p -values imply that the likelihood of these results due to chance is minimal, supporting the significance of these coefficients. Nevertheless, it is important to emphasize that correlation does not imply causation and other unexamined factors may influence economic growth. For instance, while studies generally find a positive correlation between agricultural research and development investment and economic development, the returns from agrarian research and development investment in Bangladesh may differ significantly due to low absorptive capacity for generated knowledge.

The regression results presented in Table 5 show the relationship between innovation-related variables and gross domestic product per capita growth in Bangladesh. Since the variables are expressed in levels rather than logarithmic form, the estimated coefficients should be interpreted as unit changes in the dependent variable, not as percentage changes. The results indicate that the number of citable journal articles has a positive and statistically significant effect on gross domestic product per capita growth ($\beta = 0.036486$, $p < 0.01$). This suggests that an increase in research output is associated with higher economic growth, reflecting the role of knowledge creation and dissemination in improving productivity and supporting innovation-driven development. Similarly, the number of patent applications shows a positive and significant relationship with economic growth ($\beta = 5.741583$, $p < 0.05$). This implies that applied innovation and technological advancement contribute meaningfully to economic performance by enabling the development of new technologies and improving production efficiency. The coefficient for medium and high-technology exports is also positive and statistically significant ($\beta = 20.88898$, $p < 0.01$). This highlights the importance of exporting technology-intensive goods as a channel through which innovation translates into economic growth. It suggests that economies that successfully commercialize and export technological products are more likely to experience sustained growth. In contrast, research and development expenditure does not have a statistically significant effect on gross domestic product per capita growth ($\beta = 133.9384$, $p > 0.05$). This finding indicates that increasing research spending alone may not be sufficient to drive economic growth unless it is effectively transformed into tangible outputs such as patents, innovations, and exportable technologies. From an economic perspective, these findings emphasize that innovation outputs and commercialization mechanisms play a more direct role in driving economic growth than innovation inputs alone. This implies that policy efforts should focus not only on increasing research and development investment but also on improving the efficiency of knowledge transfer, strengthening innovation systems, and promoting the export of high-value technological products.

Furthermore, the allocation of research and development spending in Bangladesh may often target low-productivity sectors or activities with delayed economic returns. This suggests that research and development investments in agriculture may not always align with immediate financial needs, which could explain the minimal effect on growth observed in this study. While several agricultural indicators, such as employment and technology exports, appear positively associated with economic growth, the research highlights the need for context-specific approaches to reform the research and development landscape. Tailored policies that promote technological

progress and employment in the agricultural sector may better unlock the sustainable economic growth potential in Bangladesh.

4.2 Research Limitations and Future Directions

This study provides valuable insights into the relationship between agricultural employment, export technology, and economic growth in Bangladesh. However, several limitations should be considered for future research. First, the regression models employed in this study restrict causal inference (Rathje et al., 2024). Unobserved factors may confound the observed associations. To address this limitation, future studies could benefit from methods to control for endogeneity, such as instrumental variables or experimental designs. Secondly, the study focuses on agricultural technology exports but does not explore the process of technological diffusion and adoption among local farmers. Future research could investigate how extension services, farmer education, and technology adoption contribute to increased productivity and poverty reduction. Studies have highlighted the role of agricultural extension services in fostering technology adoption and improving rural livelihoods (Becerra-Encinales et al., 2024). Thus, considering these factors would provide a more comprehensive understanding of agrarian innovation's impact on poverty alleviation. While this study explores a broad range of agricultural technologies, the proprietary nature of specific innovations can vary by sector. Therefore, future research should examine industry-specific technologies to provide policy recommendations tailored to particular industries, as different sectors may require unique technological advancements to enhance productivity (Hamdouna & Khmelyarchuk, 2025).

Additionally, the study uses government research and development spending as a key explanatory variable, but it does not address research and development efficiency (Li & Liu, 2025). Other potential research topics include examining the impact of successful technology transfers, patents, and innovations that emerge from public research institutions. Investigating research and development efficiency could provide valuable insights into how investments can be better allocated to maximize economic growth. Furthermore, this research does not exhaustively explore the role of infrastructure and institutional support in agricultural development. Future studies could examine how improvements in infrastructure and institutional reforms influence agricultural employment and technology exports. Infrastructure improvements are crucial to the successful adoption of farming technologies and to enhancing the efficiency of agricultural production in Bangladesh (Vasavi et al., 2025).

Lastly, the role of human capital in the adoption of agricultural technology remains understudied. Future research could focus on how education and training programs impact the uptake of agricultural technologies (Amghani et al., 2025). Recent studies suggest that human capital development, particularly through education and vocational training, plays a pivotal role in facilitating the adoption of modern farming practices in rural areas.

4.3 Policy Recommendations

The government of Bangladesh must prioritize targeted investments and policy interventions to foster sustainable economic growth and alleviate poverty through agricultural employment and technology exports. Based on this study's findings, this study proposes the policy recommendations below. Agro-industrial growth and employment should be promoted by the government to continue to support agro-industrial initiatives that create employment in the agricultural sector (Bowman, 2025). This includes offering subsidies or tax incentives to businesses engaged in the processing of farm products. Programs targeting smallholder farmers for direct employment in agro-processing industries can further enhance rural incomes. Investment in technological advancements in both the research and application of technology in agriculture is crucial (Ye, 2025). Special attention should be given to technologies such as precision farming, cold storage solutions, and mobile-based advisories that can increase agricultural productivity and reduce post-harvest losses. The government should establish public-private partnerships to foster the development and export of agricultural technologies.

Strengthening agricultural extension services is critical for the adoption of agricultural technologies. Policies should aim at expanding the reach and quality of extension services to ensure that farmers, particularly in rural areas, have access to the latest knowledge on agricultural practices and technologies. These services should be digitized to enhance accessibility and coverage. Diversification of agricultural exports is an initiative that the government should focus on by diversifying agricultural exports, moving beyond raw agrarian products to value-added products (Nugroho et al., 2021). Promoting the export of high-value farm goods such as processed foods and organic products can significantly increase foreign exchange earnings and global competitiveness. As for the focus on inclusive research and development, the related spending should be directed towards areas that directly contribute to economic growth. This includes funding for research and development in high-yield crops, innovative irrigation technologies, and sustainable farming practices. Additionally, the efficiency of research and development investments should be closely monitored to ensure that the outcomes align with the nation's economic needs.

4.4 Future Obstacles and Alternative Approaches to Overcome Challenges

While the potential of agriculture and technology exports to drive economic growth in Bangladesh is significant, several obstacles remain. Overcoming these challenges requires innovative approaches and a deeper understanding of the structural causes of poverty and inequality in Bangladesh. Infrastructure limitations are one of the key barriers to agricultural development in Bangladesh, particularly in rural areas (Khan et al., 2022). They include poor road networks, limited access to electricity, and inadequate irrigation systems. A robust infrastructure development plan should be enacted to improve rural connectivity, ensure reliable energy sources, and enhance water management systems for agriculture. Institutional weaknesses, such as weak institutions and governance issues in Bangladesh, often hinder the effective implementation of agricultural policies. These include poor coordination between government agencies, corruption, and inefficiency in the allocation of resources. Future research and policy must focus on strengthening governance and institutional frameworks to support agriculture and innovation (Gardeazabal et al., 2023). This may involve decentralizing decision-making processes to address local agricultural challenges better.

As for endogenous challenges in technology adoption, the lack of technological adoption by smallholder farmers due to issues like low levels of education, limited access to financial resources, and cultural resistance to new methods remains a significant challenge. To overcome this, the government should introduce targeted subsidies for technology adoption, along with extensive farmer education programs to build the necessary skills. Environmental vulnerability in Bangladesh is highly vulnerable to the effects of climate change, which exacerbates agricultural productivity challenges (Jalal et al., 2021). Rising sea levels, flooding, and unpredictable weather patterns threaten both agricultural output and rural livelihoods. Adapting agrarian practices to climate change through the promotion of climate-resilient technologies, such as drought-resistant crops and climate-smart farming, should be central to future policies. Moreover, the government must provide financial support and insurance schemes to protect farmers from climate-related risks.

Social and cultural barriers with widely seen gender disparities and traditional social norms may prevent some rural populations, particularly women, from fully participating in agriculture and technology adoption. Overcoming these barriers requires policies that specifically target women's inclusion in agricultural processes, including providing training, access to credit, and land rights (Hiranya & Joshi, 2025). Engaging women as equal participants in agricultural innovation can enhance productivity and contribute to poverty reduction.

5. Conclusion

This research offers empirical evidence on the role of agricultural employment, technology exports, and innovation in mitigating poverty and promoting sustainable economic growth in Bangladesh. The results from the multiple linear regression models indicate that agricultural employment, technology exports, and innovations in the farming sector positively affect per capita gross domestic product growth. However, research expenditure is not a significant contributor to economic growth. The study concludes with a recommendation for the Bangladesh government to prioritize agro-industrial production and increase employment opportunities within the agricultural sector. Additionally, exporting technology and innovations in agrarian processes could further stimulate the national economy. However, it is emphasized that increasing the research budget alone may not directly boost economic development unless such investments are strategically focused on technologies with direct financial and poverty-reduction impacts. This research has important implications for Bangladesh's development strategy, particularly in the areas of rural development, innovation, and technology transfer. It underscores the need for government investment in sectors that directly affect economic growth, with agriculture being a critical component. Furthermore, the efficiency of public investments in agricultural research and development projects should be closely monitored to ensure productivity. To enhance the effectiveness of such interventions, it is essential to implement research monitoring and evaluation tools, such as the logical framework, key performance indicators, and robust data management frameworks. These instruments will help link funding sources to the quality and impact of agricultural research and innovation.

Additionally, the study highlights the importance of policies that foster collaboration between the government, universities, private companies, and farmers. These partnerships accelerate innovation and facilitate technology transfer, which in turn could contribute to agricultural growth, sustainable development, and poverty alleviation. In summary, this study contributes valuable insights into the link between agricultural labor, export technology, and economic growth in Bangladesh. It complements existing literature on poverty reduction and sustainable development. Future research could explore other factors that influence the relationship between agricultural employment, technology, and economic growth, such as infrastructure development, access to education and training, and rural market access. This would further clarify the role of agricultural innovation in poverty reduction and sustainable economic growth.

Author Contributions

Conceptualization, S.K.M.; methodology, S.K.M.; software, S.K.M. and M.A.H.; validation, S.K.M.; formal analysis, S.K.M.; investigation, S.K.M.; resources, S.K.M.; data curation, S.K.M.; writing—original draft preparation, S.K.M.; writing—review and editing, M.A.H.; visualization, S.K.M.; supervision, M.A.H.; project administration, S.K.M.; funding acquisition, S.K.M. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflict of interest.

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